The Constitutional Constant $\theta_{\mathrm{FFT}} = \frac{1}{2}$: Conjugate Symmetry of the Real Discrete Fourier Transform

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Abstract

The Discrete Fourier Transform of any real-valued finite signal satisfies an exact conjugate-mirror symmetry, $\hat{x}_{N-k} = \overline{\hat{x}_k}$: the upper half of its spectrum is completely determined by the lower half. This classical fact (Hermitian symmetry) means that the $2N$ real parameters of the spectrum of a real signal of length N span a real subspace of dimension exactly N. Within the constitutional programme we call this fraction the *constitutional constant* $\theta_{\mathrm{FFT}} = \frac{1}{2}$, and we give the elementary proof. Counted in complex coefficients rather than real parameters, the independent fraction is $(\lfloor N/2 \rfloor + 1)/N$, not $\frac{1}{2}$. We then record what the symmetry does and does not imply. The fraction of spectral energy in the upper half equals $\frac{1}{2} - (|\hat{x}_0|^2 + |\hat{x}_{N/2}|^2)/(2E)$, so it never exceeds $\frac{1}{2}$ and is a property of the data, not a measurement of θ_{FFT}. The half-spectrum of a real signal holds exactly N real numbers, so the symmetry gives no compression relative to the input. Real invertible transforms (cosine, Hartley, Haar) have no such redundancy, so the value $\frac{1}{2}$ reflects the choice of a complex representation. A proposed parallel with Khayyam's Triangle modulo a prime fails: there the proportion of vanishing entries tends to 1. This paper is the ninth in the nine-paper programme on constitutional sieve theory [14–21].

Revision note (November 2026). The title no longer claims empirical confirmation or a bridge to Khayyam's Triangle. The definition of θ_{FFT} now counts real parameters, which give exactly $\frac{1}{2}$; the coefficient count, $(\lfloor N/2 \rfloor + 1)/N$, is stated separately, and the corollary on equal energies, which had off-by-one index bounds, is restated as magnitude symmetry. The empirical table is withdrawn: its value 0.500247 is impossible, because the energy fraction never exceeds $\frac{1}{2}$ for real data, and Section 4 now states the exact identity. The propositions on complex inputs and on numerical stability were false and are replaced by correct statements; the "adversarial robustness" proposition is restated as a bound. The "2× lossless compression floor", the 2730× and 67× compression figures and the predicted-ratio formula are withdrawn as illusory or unverified. The Khayyam "redundancy duality",

the ternary-resonance conjecture and the universality conjecture are withdrawn as false; the Hausdorff dimension of the mod-3 fractal is corrected to $\log 6/\log 3$, and $\theta_W = \frac{5}{8}$ is cited as a conjecture.

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1 Introduction

Every real-valued signal carries a hidden redundancy in its frequency representation. It is not approximate, not dataset-dependent, not an engineering convenience: it is an exact mathematical fact that follows in two lines from the definition of the Discrete Fourier Transform.

Theorem 1.1 (Informal)**.** *For any real-valued signal $x \in \mathbb{R}^N$, the upper half of its DFT spectrum is the complex conjugate, reversed, of the lower half. Storing only the lower half uniquely determines the full spectrum. Of the $2N$ real parameters of the spectrum, exactly N are independent: counted in real parameters, the fraction of the spectrum that is* independently informative *is exactly $\frac{1}{2}$.*

We call this fraction θ_{FFT}. The theorem is classical. It is usually called Hermitian (or conjugate) symmetry [9]; it is the reason real data are expanded in real cosine and sine series, a practice that goes back to Euler, Lagrange, Fourier and Gauss [6], and real-input FFT algorithms [2, 13] exploit it to save roughly half the work of a complex FFT [3]. What this paper adds is modest:

1. We name the fraction, within the programme's vocabulary, a *constitutional constant*: a structural invariant of the real DFT that holds before any data are examined. The name is a label; the content is classical.
2. We record what the symmetry does and does not imply for energy measurements, complex and quantised inputs, finite precision and compression. Several claims made for it in an earlier version—an empirical confirmation and a $2\times$ compression floor—do not hold, and an "adversarial robustness" result is a restatement of the theorem; we say why.
3. We examine a proposed parallel between $\theta_{\text{FFT}} = \frac{1}{2}$ and the modular fractal geometry of Khayyam's Triangle, and show that it fails.

The practical context is FFT-based compression of real-valued weight tensors, such as the NFR (Neural Fiber Representation) pipeline under development by the author. There the symmetry is easily misread as a guaranteed $2\times$ compression. It is not one. The independent half of the spectrum of N real numbers consists of $\lfloor N/2 \rfloor + 1$ complex coefficients, of which the DC coefficient and, for even N, the Nyquist coefficient are real: that is exactly N real numbers, as many as the input. Conjugate symmetry avoids the $2\times$ inflation of storing the full complex spectrum; it does not compress the signal. Any compression must come from spectral sparsity of the data (Section 7).

1.1 Paper organisation

Section 2 establishes notation. Section 3 gives the proof. Section 4 gives an exact identity for the energy fraction of the upper half and explains why it is not an empirical test of θ_{FFT}. Section 5 treats complex inputs, integer quantisation, adversarial inputs and finite precision. Section 6 compares the triangle of binomial coefficients modulo a prime and explains why the proposed parallel fails. Section 7 states the compression implications. Section 8 positions this paper within the constitutional sieve programme.

2 Notation and Preliminaries

Let $N \in \mathbb{N}$ with $N \geq 2$. Write $[N] := \{0, 1, \ldots, N-1\}$.

Definition 2.1 (DFT)**.** The *Discrete Fourier Transform* of $x = (x_0, \ldots, x_{N-1}) \in \mathbb{C}^N$ is the vector $\hat{x} \in \mathbb{C}^N$ with

$$\hat{x}_k \;=\; \sum_{n=0}^{N-1} x_n \, e^{-2\pi i k n/N}, \qquad k \in [N].$$

Definition 2.2 (Spectral energy, energy fraction)**.** The *spectral energy* of x is $E(x) := \|\hat{x}\|^2 = \sum_{k=0}^{N-1} |\hat{x}_k|^2$. For a subset $S \subseteq [N]$, the *energy fraction* concentrated in S is

$$\rho_S(x) \;:=\; \frac{\sum_{k \in S} |\hat{x}_k|^2}{E(x)}.$$

Definition 2.3 (Constitutional constant θ_{FFT})**.** Write $L := \{0, 1, \ldots, \lfloor N/2 \rfloor\}$ for the *lower half* of the spectrum (it contains the DC component $k = 0$ and, for even N, the Nyquist component $k = N/2$), and $U := \{\lfloor N/2 \rfloor + 1, \ldots, N-1\}$ for the *upper half*. Regard $\mathbb{C}^N$ as a real vector space of dimension $2N$, and let $\mathrm{DFT}(\mathbb{R}^N) \subseteq \mathbb{C}^N$ be the set of spectra of real signals. Define

$$\theta_{\mathrm{FFT}} \;:=\; \frac{\dim_{\mathbb{R}} \mathrm{DFT}(\mathbb{R}^N)}{\dim_{\mathbb{R}} \mathbb{C}^N},$$

the fraction of the $2N$ real parameters of a spectrum that are independent; the remaining fraction $1 - \theta_{\mathrm{FFT}}$ is derivable from them. We shall prove $\theta_{\mathrm{FFT}} = \frac{1}{2}$ exactly (Theorem 3.1), so the independent and the derivable fractions are both $\frac{1}{2}$.

Remark 2.4 (Counting coefficients instead)**.** Counted in complex coefficients rather than real parameters, the fraction is not $\frac{1}{2}$. The $\lfloor N/2 \rfloor + 1$ coefficients indexed by L determine the rest, so the independent fraction of coefficients is $(\lfloor N/2 \rfloor + 1)/N$ and the derivable fraction is $|U|/N = (N - \lfloor N/2 \rfloor - 1)/N$: for example $5/8$ and $3/8$ for $N = 8$, and $16385/32768$ and $16383/32768$ for $N = 32768$. The difference from $\frac{1}{2}$ comes from the self-paired DC and Nyquist coefficients, which are real and so carry one real parameter each.

3 Proof of $\theta_{\mathrm{FFT}} = \frac{1}{2}$

Theorem 3.1 (Constitutional constant)**.** *Let $x \in \mathbb{R}^N$ (real-valued). Then for all $k \in \{1, \ldots, N-1\}$,*

$$\hat{x}_{N-k} \;=\; \overline{\hat{x}_k}.$$

Consequently, $\hat{x}_k$ for $k \in L = \{0, 1, \ldots, \lfloor N/2 \rfloor\}$ uniquely determines $\hat{x}$ entirely, $\mathrm{DFT}(\mathbb{R}^N)$ is a real subspace of $\mathbb{C}^N$ of dimension N, and the fraction of independent real parameters is exactly $\theta_{\mathrm{FFT}} = \frac{1}{2}$.

Proof. Since $x_n \in \mathbb{R}$ we have $x_n = \overline{x_n}$. Compute directly:

$$\begin{aligned}\hat{x}_{N-k} &= \sum_{n=0}^{N-1} x_n\, e^{-2\pi i(N-k)n/N} = \sum_{n=0}^{N-1} x_n\, e^{-2\pi in}\, e^{2\pi ikn/N} \\ &= \sum_{n=0}^{N-1} x_n\, e^{2\pi ikn/N} \qquad \left(e^{-2\pi in} = 1 \text{ for } n \in \mathbb{Z}\right) \\ &= \overline{\sum_{n=0}^{N-1} \overline{x_n}\, e^{-2\pi ikn/N}} = \overline{\hat{x}_k}. \end{aligned} \tag{1}$$

The index pairing $k \leftrightarrow N-k$ is a bijection on $\{1, \ldots, N-1\}$. For even N the midpoint $k = N/2$ is its own pair ($N - N/2 = N/2$), yielding a real Nyquist coefficient. The DC component $\hat{x}_0 = \sum_n x_n$ is also real.

Counting independent real degrees of freedom:

- $\hat{x}_0$: 1 real number.
- For even N: $\hat{x}_{N/2}$: 1 real number.
- Each pair $(\hat{x}_k, \hat{x}_{N-k})$ for $k = 1, \ldots, N/2-1$: 2 real numbers (real and imaginary parts of $\hat{x}_k$; the partner is then determined). There are $N/2-1$ such pairs.

Total independent real degrees of freedom (even N): $1 + 1 + 2(N/2-1) = N$.

For odd N: DC is real (1 real dof), remaining $N-1$ coefficients pair into $(N-1)/2$ conjugate pairs each contributing 2 real dofs, giving $1 + 2 \cdot (N-1)/2 = N$ independent real dofs.

In either case the vectors $X \in \mathbb{C}^N$ with $X_{(N-k) \bmod N} = \overline{X_k}$ for all k form a real subspace of dimension N, and it contains $\mathrm{DFT}(\mathbb{R}^N)$. Since the DFT is invertible, it is injective on $\mathbb{R}^N$, so $\mathrm{DFT}(\mathbb{R}^N)$ also has real dimension N and equals that subspace. The full spectrum has N complex coefficients, i.e. $2N$ real parameters, of which exactly N are independent: $\theta_{\mathrm{FFT}} = N/(2N) = \frac{1}{2}$. □

Corollary 3.2 (Magnitude symmetry)**.** *Apart from the self-paired DC and Nyquist terms, the lower and upper halves of the spectrum carry equal energy:*

$$\sum_{k=1}^{\lceil N/2 \rceil - 1} |\hat{x}_k|^2 \;=\; \sum_{k=\lfloor N/2 \rfloor + 1}^{N-1} |\hat{x}_k|^2 .$$

Proof. From Theorem 3.1, $|\hat{x}_{N-k}| = |\hat{x}_k|$ for all k, and $k \mapsto N-k$ maps $\{1, \ldots, \lceil N/2 \rceil - 1\}$ bijectively onto $\{\lfloor N/2 \rfloor + 1, \ldots, N-1\}$. □

This is magnitude symmetry, not Parseval's identity (energy conservation between x and $\hat{x}$). The DC term and, for even N, the Nyquist term appear on neither side.

Remark 3.3 (Independence from the data)**.** The value $\theta_{\mathrm{FFT}} = \frac{1}{2}$ depends only on the signal being real; it carries no information about the *distribution* of energy within the independent half. A white-noise signal and a spectrally sparse neural weight tensor have the same θ_{FFT}. Whatever compression is achievable is determined by spectral sparsity—a property of the data, not of the DFT (Section 7).

4 The Energy Fraction $\theta^{\text{emp}}_{\text{FFT}}$

4.1 Definition and an exact identity

An earlier version of this paper measured θ_{FFT} "empirically" as the fraction of spectral energy in the upper half, together with a mirror-symmetry score:

$$\theta^{\text{emp}}_{\text{FFT}}(x) \;:=\; \rho_U(x) \;=\; \frac{\sum_{k>N/2} |\hat{x}_k|^2}{\sum_{k=0}^{N-1} |\hat{x}_k|^2}, \qquad \sigma_{\text{mirror}} \;:=\; 1 - \frac{\left\|(\hat{x}_{N-k} - \overline{\hat{x}_k})_{k=1}^{N-1}\right\|}{\left\|(\hat{x}_{N-k})_{k=1}^{N-1}\right\| + \epsilon}.$$

For real signals the first quantity is fixed by the DC and Nyquist coefficients.

Proposition 4.1 (Energy-fraction identity)**.** *Let $x \in \mathbb{R}^N$, $x \neq 0$, and $E = E(x)$. Then*

$$\theta^{\text{emp}}_{\text{FFT}}(x) \;=\; \frac{1}{2} \;-\; \frac{|\hat{x}_0|^2 + |\hat{x}_{N/2}|^2}{2E},$$

where the Nyquist term $|\hat{x}_{N/2}|^2$ is present only for even N. Consequently $0 \le \theta^{\text{emp}}_{\text{FFT}}(x) \le \frac{1}{2}$; the value $\frac{1}{2}$ is attained if and only if $\hat{x}_0 = 0$ and, for even N, $\hat{x}_{N/2} = 0$; and the value 0 is attained if and only if all the energy lies in the DC and Nyquist coefficients, as for a constant signal or, for even N, for $x_n = (-1)^n$.

Proof. Split $[N]$ into $\{0\}$, the Nyquist index $N/2$ (even N only), $L' := \{1, \dots, \lceil N/2 \rceil - 1\}$ and U. By Corollary 3.2 the energies $S_{L'}$ and S_U on L' and U are equal, so $E = |\hat{x}_0|^2 + |\hat{x}_{N/2}|^2 + 2S_U$, and $\theta^{\text{emp}}_{\text{FFT}} = S_U/E$ gives the identity. The remaining assertions follow at once. □

4.2 What the energy fraction does and does not measure

$\theta^{\text{emp}}_{\text{FFT}}$ is a property of the data, not a measurement of θ_{FFT}. Its deviation from $\frac{1}{2}$ is exactly the DC and Nyquist term of Proposition 4.1; it can be driven to 0 by a suitable signal; and it cannot tell real from complex input ($x = (1 + 2i)\, x_R$ with x_R real has the same $\theta^{\text{emp}}_{\text{FFT}}$ as x_R). Its closeness to $\frac{1}{2}$ therefore does not confirm Theorem 3.1, which in any case needs no empirical confirmation. The direct numerical check of the symmetry is the mirror residual $\max_k |\hat{x}_{N-k} - \overline{\hat{x}_k}| \,/\, \|\hat{x}\|$, or $1 - \sigma_{\text{mirror}}$; for a Gaussian random signal of length $N = 32768$ in double precision (NumPy), σ_{mirror} differs from 1 by about 3×10^{-16}.

4.3 The withdrawn table

An earlier version reported $\theta^{\text{emp}}_{\text{FFT}}$ for three synthetic 512×64 test matrices: "structured FFN weights" (low-rank plus Gaussian noise), "attention weights" (a product $(512 \times 16) \times (16 \times 64)$ with a sparse factor) and an i.i.d. Gaussian control. It included the value 0.500247 for the attention weights, with a deviation of 2.5×10^{-4} attributed to numerical precision. By Proposition 4.1 no value above $\frac{1}{2}$ is possible for the DFT of real data, and double-precision rounding errors are of order 10^{-16} in relative terms, so that entry cannot have come from the stated measurement. The table and its explanation are withdrawn.

Recomputing $\theta^{\text{emp}}_{\text{FFT}}$ for 200 random matrices of each of the three described types, flattened to length $N = 32768$, gives values between 0.49981 and 0.5, never above, with mean about 0.49997 for each type: a deviation of order $1/N$, consistent with the DC and Nyquist term of Proposition 4.1 for random data of mean near zero. Values slightly above $\frac{1}{2}$ (up to about 0.502) arise only if a two-dimensional FFT is computed and its spectrum split by flattened index, which is not a partition into conjugate pairs. The table's compression columns are discussed in Section 7.

5 Scope: Complex, Quantised and Adversarial Inputs, and Finite Precision

We record how the symmetry, and the energy fraction of Section 4, behave for four families of inputs.

5.1 Complex-valued inputs

Theorem 3.1 requires $x \in \mathbb{R}^N$. On all of $\mathbb{C}^N$ the DFT is a bijection, so no part of the spectrum is derivable from the rest: the independent fraction of real parameters is 1, not $\frac{1}{2}$. The energy fraction behaves as follows.

Proposition 5.1 (Complex inputs)**.** *Let $x = x_R + ix_I$ with $x_R, x_I \in \mathbb{R}^N$, and write $A := \hat{x}_R$ and $B := \hat{x}_I$, so that $\hat{x} = A + iB$. Then for $1 \le k \le N-1$,*

$$|\hat{x}_k|^2 - |\hat{x}_{N-k}|^2 \;=\; 4\,\mathrm{Im}\big(A_k\overline{B_k}\big).$$

Hence the lower and upper halves, excluding the DC and Nyquist terms, carry equal energy if and only if $\sum_{k=1}^{\lceil N/2\rceil-1}\mathrm{Im}(A_k\overline{B_k}) = 0$. For $N \ge 3$, $\theta^{\text{emp}}_{\text{FFT}}$ takes every value in $[0,1]$ on $\mathbb{C}^N$.

Proof. By Theorem 3.1, $A_{N-k} = \overline{A_k}$ and $B_{N-k} = \overline{B_k}$, so $\hat{x}_{N-k} = \overline{A_k} + i\,\overline{B_k}$. Expanding, $|\hat{x}_k|^2 = |A_k|^2 + |B_k|^2 + 2\,\mathrm{Im}(A_k\overline{B_k})$ and $|\hat{x}_{N-k}|^2 = |A_k|^2 + |B_k|^2 - 2\,\mathrm{Im}(A_k\overline{B_k})$. Summing the difference over $k = 1, \dots, \lceil N/2\rceil - 1$ gives the second claim. For the third, $x_n = a\,e^{2\pi i n/N} + b\,e^{-2\pi i n/N}$ has all its energy at $k = 1$ (lower half) and $k = N-1$ (upper half), in the proportion $|a|^2 : |b|^2$, so $\theta^{\text{emp}}_{\text{FFT}} = |b|^2/(|a|^2 + |b|^2)$. □

In particular, equal power in x_R and x_I is neither necessary nor sufficient for $\theta^{\text{emp}}_{\text{FFT}}(x) = \frac{1}{2}$: the signal $x = (1+2i)\,x_R$ has power ratio 4 but the same $\theta^{\text{emp}}_{\text{FFT}}$ as x_R, while $x_n = e^{2\pi i n/N}$ has equal powers and $\theta^{\text{emp}}_{\text{FFT}} = 0$.

Practical implication. Standard neural-network weight tensors are real-valued. Complex inputs arise in signal-processing applications where the data are genuinely complex (e.g., IQ radio samples, quantum state amplitudes).

5.2 Near-real inputs: integer quantisation

Quantised weights (INT8, INT4, binary) are real-valued but live in a discrete lattice. The proof of Theorem 3.1 is unaffected by the arithmetic constraint: $x_n \in \mathbb{Z}$ still satisfies $x_n \in \mathbb{R}$, so conjugate symmetry holds exactly.

Corollary 5.2. *For any $x \in \mathbb{Z}^N \subset \mathbb{R}^N$, $\hat{x}_{N-k} = \overline{\hat{x}_k}$ for all k, exactly.*

What quantisation can change is the *sparsity* of the spectrum, and with it any compression ratio; the symmetry itself is preserved.

5.3 Adversarial inputs

Proposition 5.3 (Adversarial inputs)**.** *$\theta_{\mathrm{FFT}} = \frac{1}{2}$ is a property of the space $\mathbb{R}^N$ of real signals, not of any particular signal, so no choice of real data changes it. The energy fraction, by contrast, can be placed anywhere in $[0, \frac{1}{2}]$ and no higher: for real $x \neq 0$, $0 \leq \theta^{\mathrm{emp}}_{\mathrm{FFT}}(x) \leq \frac{1}{2}$, and for $N \geq 3$ every value in $[0, \frac{1}{2}]$ is attained.*

Proof. The bounds are Proposition 4.1. For $x_n = a + b\cos(2\pi n/N)$ with a, b real, $\hat{x}_0 = aN$, $\hat{x}_1 = \hat{x}_{N-1} = bN/2$ and all other coefficients vanish, so $\theta^{\mathrm{emp}}_{\mathrm{FFT}} = (b^2/4)/(a^2 + b^2/2)$, which runs over $[0, \frac{1}{2}]$. □

So the "robustness" of θ_{FFT} to the data is automatic: it is the statement of Theorem 3.1, not an additional property. An adversary who controls the data controls the energy fraction, down to 0, but can never push it above $\frac{1}{2}$.

5.4 Finite-precision arithmetic

IEEE 754 double-precision floating point has machine epsilon $\epsilon_{\mathrm{mach}} \approx 2.2 \times 10^{-16}$. For the Cooley–Tukey FFT the computed spectrum $\tilde{x}$ satisfies, to first order, a normwise bound $\|\tilde{x} - \hat{x}\| \leq c \cdot \log_2 N \cdot \epsilon_{\mathrm{mach}} \cdot \|\hat{x}\|$, with a modest constant c that depends on the implementation and on the accuracy of the twiddle factors [7].

Proposition 5.4 (Numerical stability of the symmetry)**.** *Let $x \in \mathbb{R}^N$ and let $\tilde{x}$ be a computed spectrum with $\|\tilde{x} - \hat{x}\| \leq \delta \, \|\hat{x}\|$. Then*

$$\max_{1 \leq k \leq N-1} \left| \tilde{x}_{N-k} - \overline{\tilde{x}_k} \right| \;\leq\; 2\delta \, \|\hat{x}\| \, .$$

With the FFT bound above, the mirror residual is therefore $O(\log_2 N \cdot \epsilon_{\mathrm{mach}})$ relative to $\|\hat{x}\|$.

Proof. Since $\hat{x}_{N-k} = \overline{\hat{x}_k}$, $\left|\tilde{x}_{N-k} - \overline{\tilde{x}_k}\right| \leq |\tilde{x}_{N-k} - \hat{x}_{N-k}| + \left|\overline{\hat{x}_k} - \overline{\tilde{x}_k}\right| \leq 2\,\|\tilde{x} - \hat{x}\|$. □

This bound concerns the symmetry. It does not bound $\left|\theta^{\mathrm{emp}}_{\mathrm{FFT}} - \frac{1}{2}\right|$, which is not a rounding effect: by Proposition 4.1 it equals $(|\hat{x}_0|^2 + \left|\hat{x}_{N/2}\right|^2)/(2E)$ exactly, which is $\frac{1}{2}$ for a constant signal and typically of order $1/N$ for zero-mean random data.

6 Khayyam's Triangle Modulo a Prime: A Parallel That Fails

An earlier version of this paper argued that $\theta_{\mathrm{FFT}} = \frac{1}{2}$ and Khayyam's Triangle modulo a prime are instances of a single phenomenon, a "constitutional redundancy" of exactly one-half revealed by a change of frame. This section states the facts about the triangle and shows that the parallel fails.

6.1 Khayyam's Triangle modulo a prime

Recall from [16]: reducing the entries of Khayyam's Triangle modulo a prime p reveals a Sierpiński-type fractal. By Lucas's theorem [8], $\binom{n}{k} \not\equiv 0 \pmod{p}$ if and only if every base-p digit of k is $\leq$ the corresponding digit of n. The *void* positions in the fractal are exactly those where this digit condition fails. Consequently row n contains $\prod_i (n_i + 1)$ entries not divisible by p [4], and the first p^m rows contain $(p(p+1)/2)^m$ of them [10, 22]; for $p = 2$ this count is 3^m [4, 5], and the limiting set is the Sierpiński gasket [11]. The limiting fractal has Hausdorff dimension [10]

$$d_p \;=\; \frac{\log\bigl(p(p+1)/2\bigr)}{\log p},$$

so $d_2 = \log 3/\log 2 \approx 1.585$, $d_3 = \log 6/\log 3 \approx 1.631$, and d_p increases towards 2 as p grows. (An earlier version, following an error since corrected in [16], gave $d_p = \log(p+1)/\log p$, which is correct only for $p = 2$.)

The first p^m rows contain $p^m(p^m+1)/2$ entries in all, so the fraction of them not divisible by p is

$$\frac{(p(p+1)/2)^m}{p^m(p^m+1)/2} \;=\; \frac{2\left((p+1)/2\right)^m}{p^m+1} \;\longrightarrow\; 0 \qquad (m \to \infty),$$

and the fraction of entries that vanish modulo p tends to 1 for every prime p. For $p = 2$ and $m = 1, \ldots, 7$ the void fraction is $0, 0.10, 0.25, 0.40, 0.54, 0.65, 0.74$; for $p = 3$ and $m = 1, \ldots, 5$ it is $0, 0.20, 0.43, 0.61, 0.74$. Almost all binomial coefficients are divisible by p.

6.2 Why the parallel fails

There is therefore no redundancy of one-half in the triangle. The earlier claim that the void fraction tends to $\frac{1}{2}$ for $p = 2$ is false; the formula offered for it, $1 - p^{1-d_p}$, gives $1/(p+1)$ under that version's own d_p, i.e. $1/3$ for $p = 2$, and in any case the limit is 1. The "constitutional redundancy duality" proposition, its table of parallels and its proof sketch are withdrawn. The "conjugate pairing of rows k and $2^m - k$" invoked there does not exist, since those rows have different lengths; and a vanishing entry is not "derivable from the rest" in the sense in which $\hat{x}_{N-k}$ is derivable from $\hat{x}_k$, so the two "redundancies" are different notions. The genuine binomial analogue of conjugate symmetry is the row symmetry $\binom{n}{k} = \binom{n}{n-k}$, which determines each row from its first $\lfloor n/2 \rfloor + 1$ entries, for any modulus or none; it has nothing to do with Lucas's theorem or the fractal.

6.3 The prime $p = 3$

In the companion paper [19], the prime $p = 3$ is singled out because it is the only prime that ramifies in the Eisenstein integers $\mathbb{Z}[\omega]$ (where $\omega = e^{2\pi i/3}$). The fractal does not single out 3: Lucas's theorem and the formula for d_p hold for every prime. The level of distribution $\theta_W = \frac{5}{8}$ that [19] proposes for the moduli $q = 3^K$ is a conjecture, not a result.

An earlier version also proposed a "ternary resonance" conjecture: that three channels (magnitude, phase and "coherence", processed by three SIREN networks [12] in the NFR stage) are optimal for neural spectral compression, by a connection with "the minimality

of $d_3 = \log 4/\log 3$ among primes $p \geq 3$". Its premise used a wrong value of d_3 (under that wrong formula d_3 would be the largest value among primes $p \geq 3$, not the smallest); a single complex coefficient has two real parameters, not three; and no data supporting it were reported. The conjecture is withdrawn.

7 Compression Implications

7.1 What conjugate symmetry does and does not give

Let $w \in \mathbb{R}^N$ be a real-valued weight tensor, flattened. Its independent half-spectrum $(\hat{w}_k)_{k \in L}$, as returned by real-input FFT routines such as `numpy.fft.rfft`, consists of $\lfloor N/2 \rfloor + 1$ complex numbers, of which $\operatorname{Im} \hat{w}_0 = 0$ and, for even N, $\operatorname{Im} \hat{w}_{N/2} = 0$. It therefore holds exactly N real numbers, as many as w: for $N = 32768$, 16385 complex coefficients, i.e. 32768 real numbers.

Proposition 7.1 (No compression from the symmetry)**.** *For $w \in \mathbb{R}^N$, the map sending w to the real and imaginary parts of $(\hat{w}_k)_{k \in L}$, with the identically zero imaginary parts of $\hat{w}_0$ and (for even N) $\hat{w}_{N/2}$ omitted, is a linear bijection $\mathbb{R}^N \to \mathbb{R}^N$.*

Proof. The map is linear, and it is injective because by Theorem 3.1 the half-spectrum determines $\hat{w}$ and hence w. The number of real numbers it retains is $2(\lfloor N/2 \rfloor + 1) - 2 = N$ for even N and $2 \cdot \frac{N+1}{2} - 1 = N$ for odd N, so a linear injection $\mathbb{R}^N \to \mathbb{R}^N$ is a bijection. □

So conjugate symmetry avoids the 2× inflation of storing the full complex spectrum, but it gives no compression relative to the input. An earlier version decomposed the total compression of an FFT-based scheme as $\eta_{\text{total}} = \eta_{\text{const}} \times \eta_{\text{sparse}}$ with a "constitutional floor" $\eta_{\text{const}} = 2$, and stated a "compression lower bound" theorem: a lossless 2× compression for every real tensor, tight for white noise. Measured against the real input, $\eta_{\text{const}} = 1$, so $\eta_{\text{total}} = \eta_{\text{sparse}}$, and that theorem is withdrawn. Any compression of an FFT-based scheme comes from discarding or coarsely coding spectral coefficients—that is, from spectral sparsity of the data—and is lossy unless the discarded coefficients are exactly zero. (In floating point even the round trip $w \mapsto \hat{w} \mapsto w$ is exact only up to rounding error.)

7.2 Spectral sparsity of the test matrices

The only reproducible compression entry of the table withdrawn in Section 4 is the random control. For an i.i.d. Gaussian 512×64 matrix, keeping the largest 20% of the DFT coefficients (5×) retains about 52% of the energy (0.518 to 0.521 in our recomputation); for exponentially distributed coefficient powers the fraction retained is $f(1 - \ln f) \approx 0.522$ at $f = 0.2$.

The "2730× at 96% energy" reported for "structured FFN weights" could not be reproduced from the described construction. For low-rank (rank 1, 4, 8 or 16) plus Gaussian noise (standard deviation 0.1) matrices of that shape, 96% of the energy of the two-dimensional DFT needs between 19,871 and 23,767 of the 32,768 coefficients, a ratio of 1.4 to 1.6×. Low rank does not imply DFT sparsity: the two-dimensional DFT of UV^{T} is $(FU)(FV)^{\mathsf{T}}$, still of rank r but in general dense. A ratio of $2730 = \lfloor 32768/12 \rfloor$ would mean that 96%

of the energy lies in 12 of the 32,768 coefficients. The 2730× figure is therefore unverified and is withdrawn, together with the earlier explanation that gradient-descent training drives weight tensors "onto low-dimensional manifolds in frequency space", which synthetic test matrices could not have tested. The "moderate" entry for the attention weights was not a measurement.

7.3 The withdrawn predicted-ratio formula

An earlier version gave a "theorem-predicted" compression ratio $\eta_{\text{pred}}(d) = N/\lfloor N^{d/\dim} \rfloor$ and reported $\eta_{\text{pred}} \approx 2730 = \eta_{\text{act}}$ at $d \approx 0.03$. The symbol dim was never defined and no theorem was stated. For $N = 32768$ and $d = 0.03$ the formula gives 32768 for $\dim \in \{\frac{1}{2}, 1, 2\}$; the value 2730 would need $\dim \approx 0.1255$. The formula and the claimed agreement are withdrawn.

7.4 The NFR stage

The NFR (Neural Fiber Representation) stage trains three parallel SIREN networks [12] on a decomposition of the independent spectral half into magnitude, phase and a third "coherence" channel. It is under development, and no compression results for it are reported here. The figures given in an earlier version—67× on MoE expert weights (20× in public-facing claims), decomposed as 2×33.5—are unverified, and their factor 2 rested on the illusory floor of Proposition 7.1; they are withdrawn.

8 Position Within the Constitutional Sieve Programme

This paper is the ninth of the nine papers of the programme:

1. [15]: Constitutional Forcing. The formula $\theta_k = (2^k - k)/2^k$ is a postulate (a system of depth k is one in which exactly k configuration classes are eliminated), not a theorem.
2. [16]: Khayyam's Triangle; historical restoration and the mod-p fractal.
3. [18]: Spherical Cow Philosophy; the programme's methodology.
4. [19]: Eisenstein integers and cascade moduli; the level of distribution $\theta_W = \frac{5}{8}$ for the moduli 3^K is a conjecture.
5. [17]: Ramanujan's dimensional forcing; the counting formula is a postulate.
6. [21]: the Shannon–Wakil effect; a parallel with information theory.
7. [14]: Condition W3, the absence of exceptional zeros for characters modulo 3^K; it is true, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$.
8. [20]: the Lemke Oliver–Soundararajan bias and ternary structure.
9. This paper: $\theta_{\text{FFT}} = \frac{1}{2}$; conjugate symmetry of the real DFT.

The programme's organising idea is *constitutional forcing*: structural constraints that fix a constant before any data are examined. Of the constants mentioned in this paper, $\theta_{\text{FFT}} = \frac{1}{2}$ is a classical theorem (Theorem 3.1), $\theta_W = \frac{5}{8}$ is a conjecture [19], and the dimension of the mod-3 fractal is $\log 6/\log 3$. The value $\frac{1}{2}$ does not identify a depth k in the formula

$\theta_k = (2^k - k)/2^k$, since $\theta_1 = \theta_2 = \frac{1}{2}$. That these constants are shadows of a common "sieve geometry", as an earlier version put it, is a heuristic, not a result.

An earlier version also stated a "constitutional universality" conjecture: that every real-valued signal domain with a natural basis change (Fourier, modular, wavelet, p-adic) has a constant $\theta \in (0, 1)$ of its representation that is derivable from the remainder. It is false. Any invertible real transform $\mathbb{R}^N \to \mathbb{R}^N$—the discrete cosine transform, the discrete Hartley transform [1], or an orthonormal Haar wavelet transform—has no derivable coefficients: none of its N real outputs is determined by the others, so the derivable fraction is 0 (equivalently, the independent fraction in the sense of Definition 2.3 is 1, not $\frac{1}{2}$). The conjecture is withdrawn. The same examples show that the value $\frac{1}{2}$ is an artefact of representing a real signal by complex coefficients: the Hartley transform carries the same information as the DFT with no redundancy.

9 Conclusion

We have established the following:

1. $\theta_{\mathrm{FFT}} = \frac{1}{2}$, counted in real parameters, is an exact and classical theorem for all real-valued finite signals, proved directly from the DFT definition (Theorem 3.1). Counted in complex coefficients, the independent fraction is $(\lfloor N/2 \rfloor + 1)/N$ (Remark 2.4).
2. The energy fraction of the upper half satisfies $\theta_{\mathrm{FFT}}^{\mathrm{emp}} = \frac{1}{2} - (|\hat{x}_0|^2 + \left|\hat{x}_{N/2}\right|^2)/(2E) \leq \frac{1}{2}$ for real data (Proposition 4.1). It is a property of the data and does not test θ_{FFT}; the earlier empirical table, which contained an impossible value, is withdrawn.
3. θ_{FFT} does not depend on the data, so no real input can change it; the energy fraction can be placed anywhere in $[0, \frac{1}{2}]$ (Proposition 5.3).
4. There is no parallel with Khayyam's Triangle at the level of redundancy fraction: the proportion of entries divisible by p tends to 1 (Section 6).
5. Conjugate symmetry gives no compression relative to a real input: the half-spectrum holds exactly N real numbers (Proposition 7.1). Compression must come from spectral sparsity; the earlier 2730× and 67× figures are unverified and withdrawn.

The symmetry has been in the spectrum, and in the textbooks, all along.

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